The Mortgage Backed Securities Market and The Financial Crisis

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Introduction

Mortgage-backed securities (MBS) are a key part of modern financial markets and play an important role in connecting the housing market to money and capital markets. MBS helps turn individual home loans into tradable securities, helps increase liquidity, allows banks to issue more mortgages, and gives investors access to cash flows from housing assets. MBS has plenty of clear benefits, but also has some risks. This is the key when the underlying mortgages are not properly evaluated. A very important example of this was during the 2008 financial crisis, when problems within the MBS market contributed to major financial instability and impacted investors, financial institutions, and homeowners across the economy.

This paper's main focus is on how MBS are structured and examines the role they played in the crisis and the changes that were made after. Our main research question is: How do the structural features of MBS both support financial markets and create risk during periods of economic stress? This topic is important because MBS continue to make a large portion of fixed-income markets and directly influence interest rates, lending activity, and investment decisions. Due to this, their performance affects a wide range of groups, including policymakers, banks, and everyday investors.

The paper is organized as follows. First, it will explain the background and theory behind MBS and how they were developed. Then it looks at market data and behavior during the financial crisis. After that, it discusses the implications for investors and the financial system. Finally, it will be wrapped up with a conclusion to tie everything together.

Background and Theory

The mortgage-backed security market was established in the late 1970s, as a way to "decouple mortgage lending from mortgage investing (Fuster et al., 2022)." MBS transform illiquid mortgages into tradable securities. Prior to their development, the mortgage lending market was highly fragmented, as banks originated loans using depositor funds and held these mortgages on their own balance sheets (Fabozzi, 2016). This structure constrained banks' liquidity, leaving them vulnerable to funding shortages that limit their ability to finance new home purchases. It also exposed banks to yield curve risk, as banks borrow short-term funds at variable rates while earning fixed returns on long-term mortgages (Fabozzi, 2016). This decreases profit margins when short-term interest rates rise. To combat these issues, the national MBS market facilitates a continuous flow of funds by allowing commercial banks to originate and then sell mortgages to investors, which generates capital to issue new home loans and expands lending capacity (Fabozzi, 2016). Within this system, mortgage-backed securities are structured as bonds "with cash flows tied to the principal and interest payments on a pool of underlying mortgages (Fuster et al., 2022), forming the basis of mortgage securitization in secondary markets.

Mortgage securitization involves pooling a large number of individual home loans (with similar characteristics) and selling their cash flows to investors in the secondary markets (Fabozzi, 2016). Instead of keeping mortgages on their balance sheets, the credit risk is transferred to investors that receive payments derived from the value of the principal and interest (Fuster et al., 2022). Within this, MBS is divided into agency (backed by the full-faith of a government agency, such as Fannie Mae, Freddie Mac, or Ginnie Mae) or non-agency (backed by private financial institutions and are not guaranteed in value) (Fuster et al., 2022). Additionally, MBS are separated into tranches, which are groups of mortgages with different levels of risk and priority of repayment. This allows investors to choose securities with risk exposures that match their risk tolerance (Fabozzi, 2016).

From a theoretical perspective, the mortgage securitization process is an example of the principle of financial intermediaries that was discussed in class. They help with liquidity creation (convert illiquid loans to funds that can be withdrawn from) and mitigate risk distribution (pooling and diversification of securities). However, the originate-to-distribute model has exposed investors to information asymmetry and the principal-agent problem because commercial banks have greatly reduced borrower screening (Fabozzi, 2016). Additionally, the tranche and ranking structure of MBS does not tell the true level of underlying risk to investors. While these features can create a healthy and stable economy, they also expose structural vulnerabilities that are significant during periods of financial crisis.

Analysis

Moving forward, this analysis portion analyzes how theoretical risks of MBS during financial crises factor into market data and assess the different economic perspectives on the impact of securitization, which plays a key role in the operation of MBS.

A central principle of MBS is that mortgages act as ‘callable bonds’, in which borrowers are permitted to hold an option and pre-pay it at par when interest rates decline. Subsequently, this creates negative convexity, which means the price appreciation of a security is limited as rates drop due to accelerating pre-payments. Data from 2000-2021 shows this through the S-curve in Figure 2 of the MBS article, plotting Conditional Prepayment Rate (Fuster et al., 2022). The data reveals that while prepayments rise with “moneyness” – which is the difference between the average mortgage coupon and current market rates – they rarely reach 100%. Additionally, the curve behavior demonstrates the “burnout effect” where deep “in the money” pools become less elastic to further rate drops since the most sensitive borrowers have already refinanced (Fuster et al., 2022). The 2008 financial crisis is a prime example of the magnitude of impact market frictions can have, as CPR plummeted to a historic low of 10% in 2008. This sharp decline was spearheaded by “non-moneyness” factors, such as decreasing home prices and a weaker economy, preventing several borrowers from qualifying for refinancing. Subsequently, as shown in Figure 2, this shifted the S-curve downwards as equity-strapped borrowers were prevented from exercising their prepayment options despite low market rates (Fuster et al., 2022).

The 2008 financial crisis also affected the Option-Adjusted Spread (OAS), which is the main metric for assessing MBS valuations, representing the expected excess return after accounting for prepayment risk (Fuster et al., 2022). Analysis of JP Morgan data revealed that while OAS normally averages above 50 basis points, it rose sharply to 150 basis points during the 2008 crisis due to liquidity and credit concerns (Fuster et al., 2022). This proceeding is often linked to what is known as “preferred-habitat effects” or model error, which is when investors demand additional returns for taking on risks outside of their manageable comfort zone. In response to this sequence, the Fed implemented its first Large Scale Asset Purchase Program (QE1), the beginning of a decade-long Central Bank intervention in the MBS market (Fuster et al., 2022). Research confirms these announcements and purchases effectively reduced MBS yields and narrowed the OAS, showing the Federal Reserve’s ability to lower mortgage rates with direct market support (Fuster et al., 2022).

The 2008 crisis also revealed a substantial deviation in market resilience based on security design. Although the To-Be-Announced (TBA) market provided a liquid avenue for agency MBS to continue trading, the nonagency MBS experienced a “freeze” in mid-2007 (Fuster et al., 2022). Nonagency issuance screeched to a halt and has remained at pre-crisis levels for decades pos-crisis. Currently, the daily trading volume for agency residential MBS is approximately $288 billion while nonagency RMBS pales in comparison at a mere $0.5 billion. This difference shows the lasting impact of the crisis on investor trust in private-label securitization (Fuster et al., 2022).

Discussion

Based on the analysis, there could be implications that MBS could see issues from CPR drops, where people demand more for investments and a larger interest in other securities. The CPR drops were due to many reasons, but the most significant was that people defaulted on their mortgages. This led to MBS suffering as a result (Deaton, 2012). The drops would make investors start to liquidate en masse, since their MBSs could likely fail due to aforementioned defaults. However, the MBS would be stuck with them, since no one wants to invest in a failing market. This would lead into a massive collapse in other markets and institutions as well. The hardest hit would be investment banks, who would struggle with issuing new MBSs. Companies would struggle less, but with the MBS market's large size (with around $100 billion issued per month in the modern market), companies would have heavy investments in MBSs (US Mortgage, 2026). With many MBSs vanishing, companies would try to lay off workers to stay afloat during the crisis. This leads to people not only walking away from their houses, but them losing their jobs as well (Deaton, 2012). Lawmakers should consider trying to implement more proactive, rather than reactive, measures to avoid a similar situation.

The failure of MBSs would lead to investors demanding more from MBSs to compensate for the risk. Even after the crisis is over, investors would likely still keep the higher rates and would suffer MBS demand issues. This is because despite the real risk of a MBS failing lowering after the crisis, investors would have the crisis stuck in their mind leading to the perceived risk being much higher. Demand for MBSs would also be stifled, since people need time to build up capital for a mortgage and hedgers would likely avoid MBSs for a time. The perceived risk could lead to commercial banks being stuck with mortgages since there would not be enough demand for MBSs. This is supported by relatively slow growth after the financial crisis (Treasury, 2026).

After the crisis, other securities could be partly helped from MBSs falling. Since MBSs would be perceived as too risky, other securities could see some growth as investors jump ships. It is possible that other securities fared better than MBSs, but the 2008 crisis was so severe that any benefits from investors purchasing other securities paled in comparison to the severe decline the securities experienced (Yahoo!, 2026). After the crisis though, other securities would likely recover faster due to investors avoiding MBSs for reasons listed in the paragraph above. It is unclear if other securities were able to substantially recover faster than the MBSs market with the stock market having similar slow growth after 2009 (Treasury, 2026; Yahoo!, 2026).

Conclusion

Ultimately, it is clear that mortgage backed securities provide investors with a new class of investments that can be tailored to meet specific demands, such as level of risk or repayment structure. They play a central role in the modern market by transforming illiquid mortgage loans into a tradable asset, expanding the lending capacity for commercial banks. However, the 2008 crisis revealed the dual role that MBS plays: the same structural features that make the securities effective also create increased risk. Most substantially, this includes information asymmetry issues and weakened lending standards that shows the systemic instability of MBS. Since then, there has been a new policy that introduced regulatory oversight, targeting to improve transparency, strengthen bank underwriting, and restore investor confidences. Thus, policymakers and investors can use the 2008 crisis to navigate future economic crises.

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